Innovative Science Olympiad

2026-2027

Grade 8

Total Questions: 35

LEVEL – II

Set - B

Time: 1.5 hr.

Total Marks: 70

SECTION - 1	• This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 2	• This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s). • For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. • There are **NO NEGATIVE** marks for wrong answers.
SECTION - 3	• This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. • Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 4	• This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS.** • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A liquid is filled in two containers of different shapes but same height. Pressure at the bottom is:

(A) Greater in wider container
(B) Greater in narrower container
(C) Same in both
(D) Depends on volume

2. Gallium has a melting point of approximately 29.7°C. If you held a piece of Gallium in your hand (average body temperature 37°C), what physical change would occur?

(A)It would melt into a liquid due to the transfer of thermal energy from your body.
(B)It would emit a bright glow as it reaches its 'incandescence point'.
(C)It would sublimate directly into a gas.
(D)It would become harder and more brittle due to the salt on your skin.

3. **A cell is often compared to a factory. If the "Nucleus" is the Manager's Office, which of the following comparisons is INCORRECT?**

(A) Cell Membrane — Security Guard (controls entry and exit).
(B) Mitochondria — Power Plant (generates energy).
(C) Vacuoles — Storage Room (stores waste or food).
(D) Cytoplasm — The finished product being sold (the final output).

4. **A block of mass 10 kg lies on a rough horizontal surface with coefficient of static friction μ = 0.5. A force F is applied at an angle 30° downward with the horizontal (pushing). What is the minimum force required to just start motion? (Take $g = 10\ m/s^2$)**

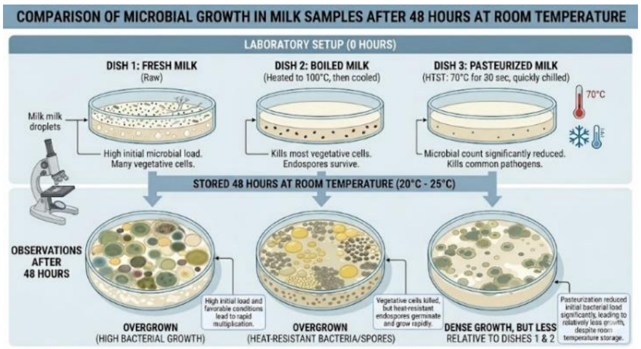
(A) 40 N (B) 50 N
(C) 66.7 N (D) 81 N

5. **Why is Sodium highly reactive with water, while Magnesium only reacts with hot water/steam, and Iron only reacts with steam at red-hot temperatures?**

(A)Iron atoms are larger than Sodium atoms.
(B)It is due to the decreasing ionization enthalpy as we move from Sodium to Iron.
(C)The chemical reactivity decreases as the ease of losing valence electrons decreases across the series.
(D)Water behaves as an acid with Sodium but as a base with Iron.

6. **In a laboratory, three Petri dishes were set up to test food preservation. Dish 1 had fresh milk, Dish 2 had boiled milk, and Dish 3 had "Pasteurized" milk (heated to 70°C for 30 seconds and then chilled quickly). After 48 hours at room temperature, which dish will likely show the least microbial growth, and why?**

(A) Dish 1; fresh milk has natural antibodies that kill all bacteria.
(B) Dish 2; boiling kills every single microorganism including heat-resistant spores.
(C) Dish 3; Pasteurization kills most harmful microbes without destroying the nutritional quality of the milk.
(D) All dishes will have the same growth because milk is a natural medium for bacteria.



7. **A tuning fork vibrating at 250 Hzgenerates sound in the air (speed ≈ 340 m/s). The sound waves then cross into a pool of water (speed ≈ 1500 m/s). What correctly describes the changes to the frequency and wavelength of the wave as it enters the water?**

(A) Frequency increases to match the new speed; wavelength remains constant.
(B) Both frequency and wavelength increase proportionally due to the denser medium.

(C) Frequency remains exactly 250 Hz; wavelength increases.
(D) Frequency decreases; wavelength decreases to conserve wave energy

8. **A scientist finds that a coal sample has a very high moisture content and a low carbon percentage (around 25-35%). To which stage of coal formation does this sample likely belong?**

(A) Peat (B) Anthracite
(C)Lignite (D) Bituminous

9. **A farmer has a 5-hectare field. He realizes that using a traditional "Broadcasting" method (scattering seeds by hand) results in 30% of his seeds being eaten by birds and 20% failing to grow due to overcrowding. If he switches to a "Seed Drill," which sows seeds at a depth of 5 cm and at uniform distances, how does this specifically solve his problem?**

(A) The depth prevents birds from reaching the seeds, and the uniform spacing reduces competition for nutrients and sunlight.
(B) The seed drill adds chemical fertilizers automatically, which makes the seeds taste bitter to birds.
(C) Using a machine increases the soil temperature, which speeds up germination by 50%.
(D) The seed drill removes all weeds automatically while it is sowing the seeds.

10. **A force *F* is applied such that block is just about to move. If force is suddenly reduced slightly, friction:**

(A) Becomes zero
(B) Remains same
(C) Becomes equal to new force
(D) Becomes constant

11. **Why is water ineffective at extinguishing a fire caused by a short circuit or burning petrol, but effective for a pile of wood?**
I. Water conducts electricity, posing a risk of electrocution.
II. Water is denser than petrol, causing the oil to float and continue burning.
III. Water raises the ignition temperature of wood chemically.
IV. Water cools the wood below its ignition temperature and cuts off oxygen via steam.

(A)I and II only (B) I, II, and IV
(C) II and III only (D) II only

12. **An ecosystem in a "Buffer Zone" of a Biosphere Reserve is experiencing a severe drought. The local government decides to plant thousands of eucalyptus trees (a fast-growing non-native species) to quickly restore the green cover. Why might conservationists argue against this "quick fix"?**

(A) Eucalyptus trees produce too much oxygen, which can be toxic to local insects.
(B) Introducing non-native species can disrupt the local food chain and suck up excessive groundwater, harming native biodiversity.
(C) Non-native species are not capable of performing photosynthesis in a Biosphere Reserve.
(D) Eucalyptus trees attract too many migratory birds, which creates noise

pollution for the animals in the National Park.

13. **In male humans, the testes are located outside the abdominal cavity in a sac called the scrotum. What is the biological reasoning for this specific anatomical placement?**

(A) To protect the testes from being crushed by the internal organs.
(B) To keep the temperature of the testes 2-3°C lower than the body temperature, which is necessary for sperm production.
(C) To allow the sperm to travel faster into the lungs.
(D) Because the testes are not part of the reproductive system.

14. **During the study of animal reproduction, a student compares the "Asexual Budding" of *Hydra* with the "Sexual Reproduction" of a human. What is the fundamental genetic difference between the offspring produced in these two processes?**

(A) The *Hydra* offspring has more genetic variation because it grows directly from the parent's body.
(B) The human offspring is a genetic clone of the father only.
(C) The *Hydra* offspring is genetically identical to its single parent, while the human offspring has a unique combination of genes from two parents.
(D) Both methods produce offspring that are 100% identical to the mother.

15. **A student views a cell under a microscope and observes a large, dark, spherical body in the center, but no distinct cell wall. He also sees many small, hair-like structures on the outside of the cell membrane used for movement. What type of cell is this most likely to be?**

(A) A plant cell from a leaf.
(B) A bacterial cell.
(C) An animal cell
(D) A fungal cell from a mushroom.

Section - 2

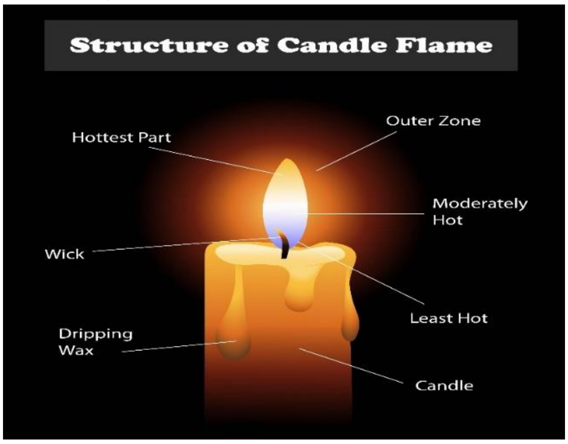
16. **In a liquid, the pressure at a depth of 3 m is 36 Pa. Assume the liquid is at rest and density is uniform. Which of the following statements are correct?**

(A) Pressure at 6 m depth is 72 Pa
(B) Pressure difference between 6 m and 3 m is 36 Pa
(C) Pressure at 1 m depth is 12 Pa
(D) Pressure at a given depth depends on the shape of the container

17. **A student is observing a steady candle flame. Which of the following statements regarding the zones and properties of this flame are correct?**

(A)The innermost zone is the hottest because it contains unburnt wax vapors that act as concentrated fuel.
(B)The middle zone appears yellow due to the presence of unburnt carbon particles that glow (incandescence).
(C)In the outermost zone, complete combustion occurs because there is a plentiful supply of oxygen.
(D) A copper wire held in the luminous zone will get blackened, while a glass slide

held in the non-luminous zone will remain relatively clear



18. "Reforestation" is the restocking of destroyed forests by planting new trees. For an effective and ecologically sound reforestation program, which of the following should be considered?

(A) The trees planted should generally be of the same species that were originally found in that forest.
(B) We should plant at least as many trees as we cut down to maintain balance.
(C) Natural reforestation can happen if the deforested area is left undisturbed for a long time.
(D) Planting only exotic ornamental plants is the best way to restore a native forest ecosystem

19. A person pushes a box inside a moving train with constant force F.

(A) Work done in train frame is $F \cdot s$
B) Work done in ground frame is $F \cdot (s + s_0)$
(C) Work done is same in all frames
(D) Work depends on displacement, which can vary with frame

20. Spontaneous combustion is a specific type of burning. Which of the following scenarios or statements accurately describe this phenomenon?

(A) It occurs without any external application of heat because the substance's ignition temperature is reached through internal slow oxidation.
(B) The burning of phosphorus at room temperature is an example of spontaneous combustion.
(C)Explosion and spontaneous combustion are identical because both happen very rapidly.
(D) Forest fires can sometimes be caused by spontaneous combustion due to the heat of the sun or lightning strikes.

21. "Mixed Cropping" and "Intercropping" are often confused. Which of the following statements correctly describe these practices?

(A) Mixed cropping involves sowing two or more crops simultaneously on the same piece of land without a set pattern.
(B) Intercropping involves growing two or more crops simultaneously on the same field in a definite row pattern.
(C) Both practices help in reducing the risk of total crop failure due to pests or weather.
(D) Both practices always require the use of different tractors for each row of crops.

22. Which of the following diseases are correctly matched with their causative "Pathogen" and "Mode of Transmission"?

(A) Tuberculosis — Bacteria — Air-borne.
(B) Malaria — Virus — Female *Anopheles* mosquito.
(C) Citrus Canker — Bacteria — Air-borne.

(D) Rust of Wheat — Fungi — Air and Seeds.

23. An "Endemic Species" is at a higher risk of extinction than a cosmopolitan species. Why?

(A) Their population is restricted to a very specific geographical area.
(B) Any disturbance to their specific habitat (like a forest fire or local pollution) can wipe out the entire species.
(C) They are unable to adapt to any environment other than their native them vulnerable.
(D) They are always predators at the top of the food chain, which makes them vulnerable.

Section - 3

COMPREHENSION-1

Two wires A (copper) and B (nichrome) of identical length and thickness are connected separately to the same battery. Wire B glows while A does not. A compass is placed near both wires.

24. The reason B glows more is:

(A) Higher conductivity
(B) Lower resistance
(C) Higher resistance causing more heat
(D) Stronger magnetic field

25. If same current flows in both wires, magnetic effect near them is:

(A) Greater in B
(B) Greater in A
(C) Same in both
(D) Zero in A

26. If length of wire B is doubled (same material), heat produced will:

(A) Increase
(B) Decrease
(C) Remain same
(D) Become zero

COMPREHENSION-2

In a modern refinery, simple distillation is not enough. The market demand for **Gasoline (Petrol)** is much higher than the amount naturally found in crude oil. Conversely, there is often an excess of heavier, less valuable fractions like **Fuel Oil**.

To balance this, refineries use a process called **Catalytic Cracking**. In this process, large hydrocarbon molecules (C15–C25) are "cracked" into smaller, more useful molecules (C5–C10) using high temperatures and a **zeolite catalyst**. This process not only increases gasoline yield but also produces **Alkenes** (like ethene and propene), which are the essential raw materials for the plastics industry.

27. A chemical engineer is monitoring a cracking unit. The input is a heavy gas oil fraction ($C_{20}H_{42}$). After the reaction, the output contains a mixture of C_8H_{18} (Octane) and another hydrocarbon X. Based on the law of conservation of mass, what is the most likely identity of X?

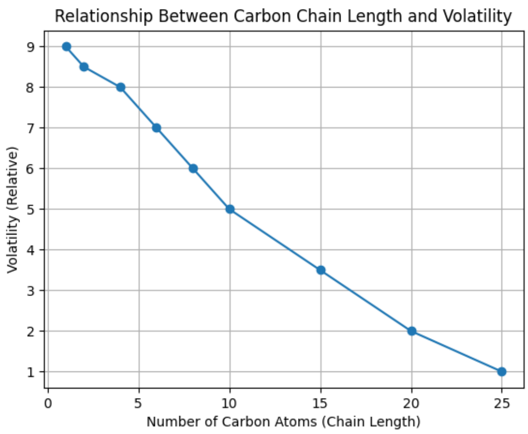
(A) Two molecules of C_6H_{12}
(B) One molecule of $C_{12}H_{24}$
(C) Three molecules of C_4H_8
(D) One molecule of $C_{10}H_{22}$

28. Why is a catalyst preferred over purely thermal cracking (using only heat) in modern refineries?

(A) The catalyst chemically bonds with the carbon to create "greener" fuel.

(B) The catalyst lowers the activation energy, allowing the reaction to occur at lower temperatures, saving energy.
(C) The catalyst increases the boiling point of the final gasoline, making it safer to store.
(D) The catalyst prevents the formation of any gaseous by-products.

29. The graph below shows the relationship between the length of a carbon chain and its volatility (how easily it turns into gas).



Based on this relationship, which of the following is a logical deduction regarding the safety of refinery storage?

(A) Bitumen is more likely to cause an explosion at room temperature than Petrol.
(B) Storing Refinery Gases (C1-C4) requires high-pressure tanks because their volatility is extremely high.
(C) As the number of carbon atoms increases, the risk of the fuel evaporating spontaneously increases.
(D) There is no relationship between chain length and storage requirements

COMPREHENSION-3

During a summer camp, students noticed that cooked rice left outside for a long time developed a foul smell. The teacher explained that microorganisms grow rapidly in warm and moist conditions, leading to food spoilage.

In another situation, milk stored in the refrigerator remained fresh for a longer time compared to milk kept at room temperature. The teacher said that low temperature slows down the growth of microorganisms.

Later, the students learned that some microorganisms are used in making bread. Yeast helps the dough rise by producing gas during fermentation.

30. Why did the cooked rice develop a foul smell?

(A) Loss of nutrients
(B) Growth of microorganisms
(C) Cooling of rice
(D) Exposure to sunlight

31. What is the main reason microorganisms grow faster in summer?

(A) Low humidity
(B) Warm temperature
(C) Less oxygen
(D) Dry conditions

32. Which of the following methods is best to prevent food spoilage?

(A) Keeping food in warm places
(B) Storing food in moist conditions
(C) Refrigeration
(D) Exposing food to air

Section - 4

33.

COLUMN A	COLUMN B
(A) Image disappears when pinhole is enlarged	(P) Rays fail to intersect at a point
(B) Two mirrors inclined give multiple images	(Q) Multiple reflections due to angle
(C) Object seen behind actual position in glass slab	(R) Lateral shift due to refraction
(D) Light deviates but returns parallel after slab	(S) Emergent ray property

(A) A–Q, b–P, c–R, d–S
(B) A–P, b–Q, c–R, d–S
(C) A–Q, b–R, c–P, d–S
(D) A–R, b–P, c–Q, d–S

34.

COLUMN -A	COLUMN-B
(A) Graphite	(P) Liquid at room temperature, used in thermometers
(B) Mercury	(Q) Non-metal used as an electrode due to conductivity
(C) Phosphorus	(R) Stored in kerosene to prevent spontaneous combustion
(D) Sodium)	(S) Stored under water to prevent catching fire in air

(A) A–R, B–P, C–Q, D–S
(B) A–P, B–R, C–S, D–Q
(C) A–R, B–P, C–S, D–Q
(D) A–Q, B–P, C–S, D–R

35.

COLUMN -A	COLUMN-B
(A) Cytoplasm	(P) Storage of substances
(B) Vacuole	(Q) Site of chemical reactions
(C) Chloroplast	(R) Photosynthesis
(D) Cell wall	(S) Provides support

(A) A – P, B – Q, C – R, D – S
(B) A – Q, B – P, C – R, D – S
(C) A – R, B – S, C – Q, D – P
(D) A – Q, B – P, C – R, D – S